Customer Lifetime Value (CLV) Prediction and Retention Budget Optimization Using Machine Learning

Subhadra Mishra

Background / Problem

Telcos lose revenue from churn. Retention campaigns often waste budget on low-risk customers.

- Predict churn probability
- Estimate Expected Loss (P(churn) × CLV)
- Optimize retention spend under budget
- Compare targeted vs. random strategies

Methods and Data

1. Preprocessing: Load data, convert TotalCharges, map Churn to binary, one-hot encode.
2. Modeling: Logistic Regression + Random Forest; ROC-AUC, precision, recall.
3. Simulation: Expected Loss = P(churn)×CLV; targeted vs. random under fixed budget.
4. Quantification: Revenue saved, ROI, targeting efficiency.

Results

Model: ROC-AUC 0.851 | Precision 0.552 | Recall 0.735

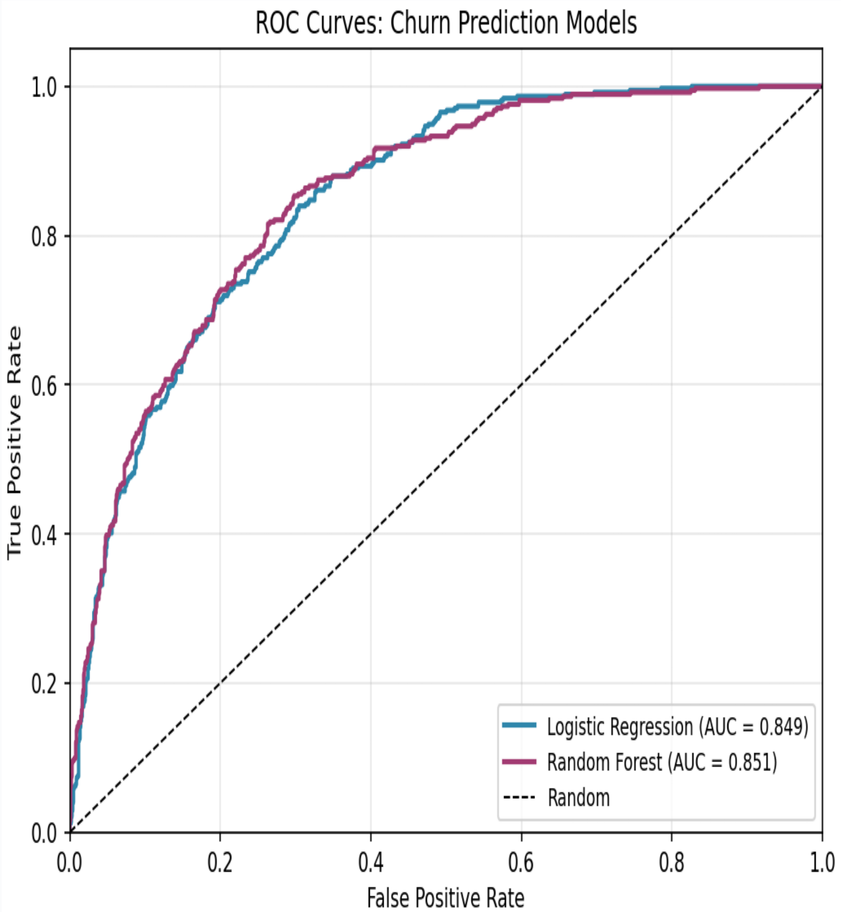
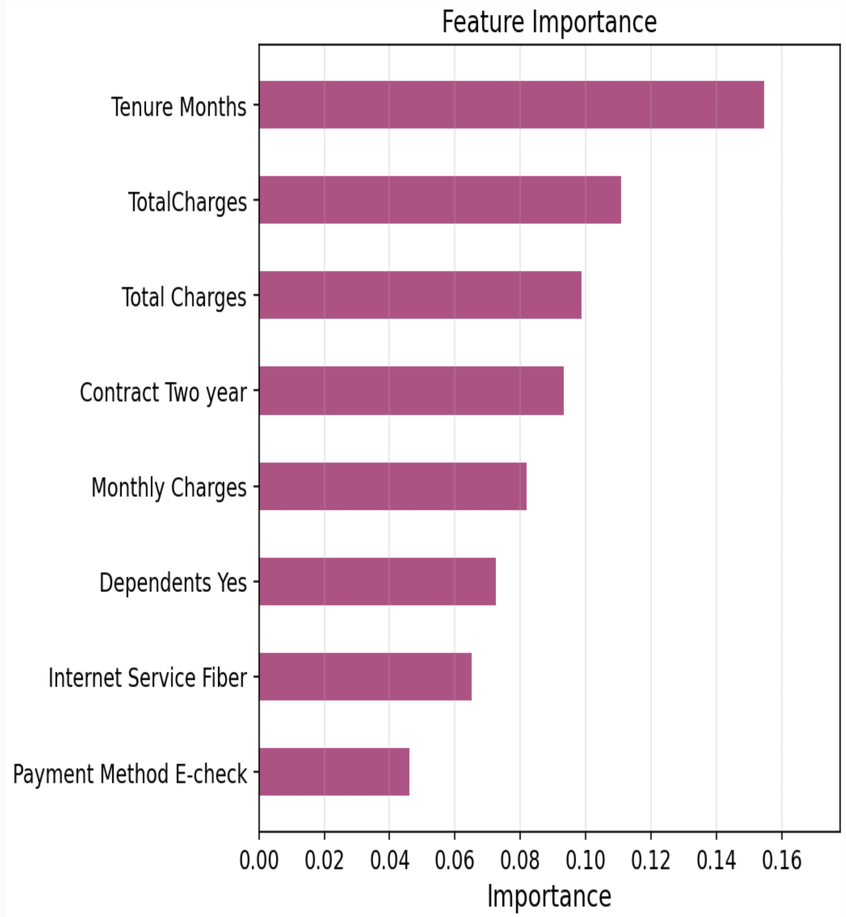
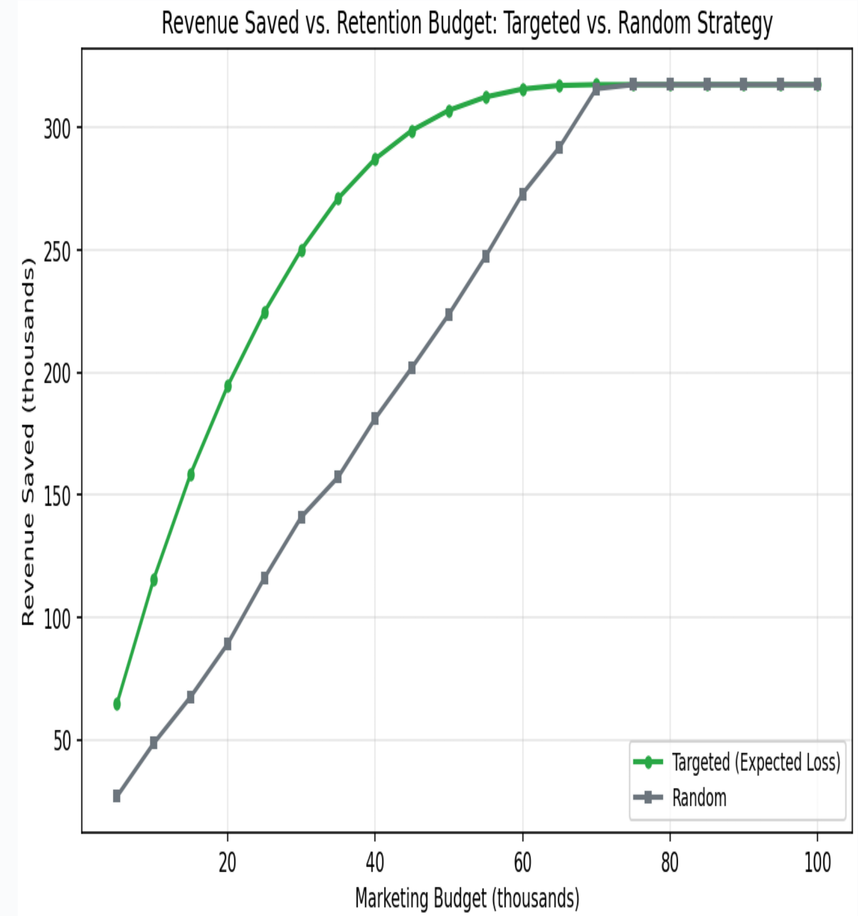
Simulation (25*Kbudget*, 50/customer, 15% reduction):
- Targeted: $224,684 saved | ROI 799%
- Random: $116,083 saved | ROI 364%
- Efficiency: 1.94×

Sensitivity: robust across 5–25% churn reduction, 25–100 cost.

References

IBM Telco Churn Dataset
- github.com/IBM/telco-customer-churn-on-icp4d
- Kaggle: waseemalastal/telco-customer-churn-ibm-dataset
- License: Apache-2.0 | 7,043 customers, 31 features

Analysis / Conclusions

- Rank by Expected Loss → maximize revenue saved under budget
- Targeted strategy ~2× more revenue than random
- Sensitivity + budget scaling: results robust
- Reproducible pipeline: preprocessing, modeling, simulation

Key takeaway: Budget scaling shows efficiency holds across 5*K*–100K. Top drivers: tenure, monthly charges, contract type.